

FDE Xlerate

Week 1, Day 2

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The problem

■ Task

- Build the agent loop by hand with no framework.
- With no framework (e.g. LangChain, LangGraph, or CrewAI), each component of the agentic loop is made clear.

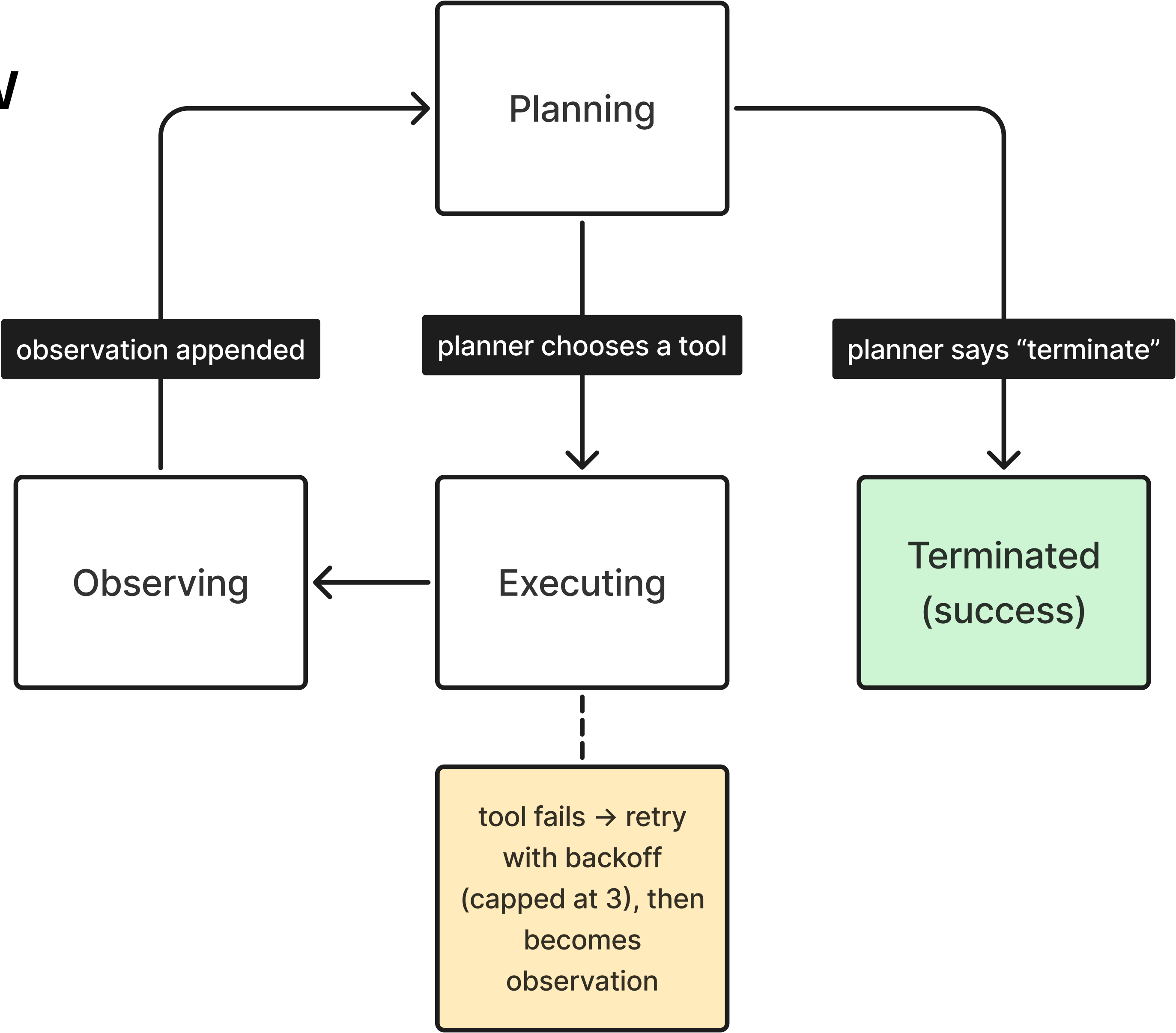
■ Business problem

- Industry: insurance
- Case: Investigate a submitted insurance claim for fraud signals before it reaches a human adjuster.

When an insurance claim is made, it must be investigated for fraud signals to ensure legitimacy. Fraud signals may be spread across multiple data sources and any decision may rely on related information between sources. For example, claim history and max payout from a policy document may combine to indicate repeated instances of near-max claims.

A fraud investigation cannot be scripted as a workflow as decisions (conclusion made, more evidence needed, what evidence may support evidence-based hypothesis, etc.) must be made dynamically against each new piece of evidence pulled.

High-level flow



Architecture

■ Tool registry

registry.py

- name → callable + JSON-schem-style args + description
- 4 tools
 - get_policy_details
 - get_claim_history
 - check_repair_shop_reputation

■ Planner

planner.py

- One Claude call per turn, constrained to two JSON output shapes (call_tool or terminate).
- Anything else is treated as a harness-visible contract violation and raised.

■ Termination checker

- Runs before the planner is called in each iteration so that the loop's upper bound doesn't depend on the planner or any particular tool cooperating

■ Agent state

state.py

- one dataclass threaded through the whole run: goal, observation history, iteration/ token/ time counters, status
- No module-level global - a run can be dumped to JSON mid-flight

■ Executor

executor.py

- Dispatches the tool, owns exponential backoff with jitter (capped at 3 attempts), and always returns an observation
- A tool failure becomes data for the planner, never a crash

Tech Stack

■ No agent framework

Python for the loop registry, and executor

■ No persistent memory

In-memory state (a dataclass) with no persistence

■ Unit tests

39 unit tests covering state invariants, each tool, the retry/backoff mechanics, planner-contract validation, and the loop's termination guarantees

■ Model

Claude Sonnet 5 via the Anthropic API as the planner

■ Backoff

Hand-written backoff: $\text{delay} = 0.5 * 2^{\text{attempt}} + \text{jitter}$